

Quantum Communication

An Introduction

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Quantum Communication

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1 Review of Quantum Computing

Quantum computers have the potential to accelerate certain computational tasks that are intractable for classical systems. At the same time, they pose a fundamental threat to current cryptographic standards.

Quantum mechanics foundations: *Superposition*, *entanglement*, and *measurement*.

Quantum computers: Unlike classical bits, they operate on *qubits*.

Quantum algorithms: They use constructive interference to amplify the probability of the correct solution while canceling wrong ones.

NISQ era: Hardware is prone to decoherence and lacks error correction.

FTQC era: Executing deep circuits required for practical applications.

Cryptographically Relevant Quantum Computer (CRQC): quantum processor capable of executing algorithms like Shor's and Grover's.

2 Review of Post-Quantum Cryptography

Due to the cryptographic threat posed by CRQC, governments and regulatory bodies are urging private organizations and public institutions to transition to quantum-safe cryptography.

Cryptography is the “art and science” of securing information.

Difficult mathematical problems underlying current cryptographic standards: *RSA* (Rivest–Shamir–Adleman), *Elliptic Curve Cryptography* (ECC), *Diffie–Hellman* (DH), *Digital Signature Algorithm* (DSA).

Asymmetric (Public Key) Cryptography: RSA, ECC, DSA, DH. Total collapse via Shor’s algorithm (exponential speedup).

Symmetric (Secret Key) Cryptography: AES-128, AES-256. Weakened via Grover’s algorithm (quadratic speedup).

The main obstacle to implementing Shor’s algorithm is T -gate cost.

The *Harvest Now, Decrypt Later* (HNDL) scenario.

A *cryptographic primitive* is a basic mathematical procedure that performs a specific security function.

A *protocol* is a sequence of steps that specifies how multiple cryptographic primitives are combined and used to achieve secure communication between parties.

TLS example:

Digital signatures enable a server to prove its authenticity and the integrity of messages using a private key.

Key exchange allows two parties to establish a shared secret over an insecure channel.

Encryption transforms data into an unreadable form to ensure confidentiality.

Quantum attacks primarily target the asymmetric “handshake”, not the symmetric encryption.

Post-Quantum Cryptography (PQC) is based on new classes of mathematical problems believed to be hard even for quantum computers.

Increasing regulatory and legislative pressure is driving organizations to begin migrating toward PQC solutions (FIPS 203–205, 206 expected).

This is not a simple upgrade, but a fundamental transformation of digital infrastructure, strategic priorities, and organizational culture.

Crypto agility is the ability to adapt to evolving cryptographic requirements in a fast and efficient manner.

Hybrid solutions act as a practical bridge, enabling a gradual and secure transition toward fully post-quantum systems.

Alternative approaches, such as *Quantum Key Distribution (QKD)*, use the laws of quantum mechanics to establish secure communication channels.

3 Entanglement-Based Communication

Quantum entanglement enables communication protocols that have no classical counterpart.

3.1 Entanglement

Entangled states are the fundamental building blocks of quantum communication.

Quantum mechanics was developed in the late 1920s.

In 1932, John von Neumann formalized its mathematical framework in *Mathematical Foundations of Quantum Mechanics*.

In 1935, Einstein, Podolsky, and Rosen published the famous *EPR paradox* paper: *On the Completeness of Quantum-Mechanical Description of Physical Reality*.

EPR argued that quantum mechanics appeared to be incompatible with classical notions of locality and realism.

In 1964, John Bell derived inequalities that made these questions experimentally testable.

Subsequent experiments (notably by Alain Aspect in the 1980s) showed violations of Bell inequalities, confirming the non-local correlations predicted by quantum mechanics.

These results established entanglement as a real physical phenomenon and a key resource for quantum communication.

The first QKD protocol was introduced by Bennett and Brassard in 1984 (BB84). In 1991, Ekert proposed an entanglement-based approach (E91).

Together, these protocols define the foundations of Quantum Key Distribution (QKD). QKD is part of quantum cryptography, enabling secure communication based on the laws of physics.

3.2 The Math of Entangled States

We define a *single qubit* (1 qubit) as

$$|q_1\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$$

where $|0\rangle, |1\rangle$ are two orthogonal vectors and α_0, α_1 are in general complex numbers. A *2 qubit* is defined as

$$|q_2\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

Entangled state: a state that cannot be written as the tensor product of two vector states. For an entangled 2 qubit

$$\begin{aligned} |q_2\rangle &\neq |q\rangle \otimes |q'\rangle \\ &= (\alpha_0|0\rangle + \alpha_1|1\rangle) \otimes (\alpha'_0|0\rangle + \alpha'_1|1\rangle) \\ &= \alpha_0\alpha'_0|00\rangle + \alpha_0\alpha'_1|01\rangle + \alpha_1\alpha'_0|10\rangle + \alpha_1\alpha'_1|11\rangle \end{aligned}$$

There are correlations that cannot be explained by classical physics.

3.3 Bell (EPR) states

The state

$$\beta_0 = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

is an entangled state.

Proof: Suppose it is not entangled

$$\alpha_{00} = \alpha_0 \alpha'_0 \quad \alpha_{01} = \alpha_0 \alpha'_1 \quad \alpha_{10} = \alpha_1 \alpha'_0 \quad \alpha_{11} = \alpha_1 \alpha'_1$$

that is

$$\frac{1}{\sqrt{2}} = \alpha_0 \alpha'_0 \quad 0 = \alpha_0 \alpha'_1 \quad 0 = \alpha_1 \alpha'_0 \quad \frac{1}{\sqrt{2}} = \alpha_1 \alpha'_1$$

which is an inconsistent set of relations. Thus, $|\beta_0\rangle$ is entangled. $\square$

Other Bell states ($i,j=0,1$)

$$|\beta_{ij}\rangle = \frac{1}{\sqrt{2}}(|0i\rangle + (-1)^j |1\bar{i}\rangle)$$

3.4 The Hadamard and CNOT Gates

Circuit representation of the *Hadamard gate*

$$|q\rangle \longrightarrow \boxed{H} \longrightarrow H|q\rangle$$

It is defined by

$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle \quad H|1\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = |-\rangle$$

The *CNOT gate* is a 2-qubit gate

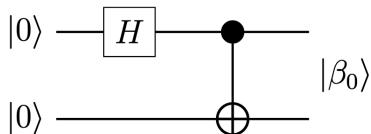


defined by (for $i, j \in \{0, 1\}$)

$$|i\rangle|j\rangle \xrightarrow{\text{CNOT}} |i\rangle|j \oplus i\rangle$$

3.5 Generating Bell States

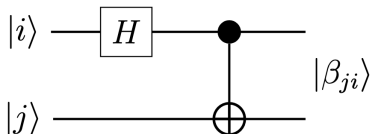
For the state $|\beta_0\rangle$



In fact

$$|0\rangle|0\rangle \xrightarrow{H \otimes I} |+\rangle|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Similarly, for the other Bell pairs



$$|01\rangle \xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle \xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |00\rangle$$

$$|10\rangle \xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|1\rangle \xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = |10\rangle$$

$$|01\rangle \xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|0\rangle \xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |00\rangle$$

$$|11\rangle \xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|1\rangle \xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = |10\rangle$$

Using index notation

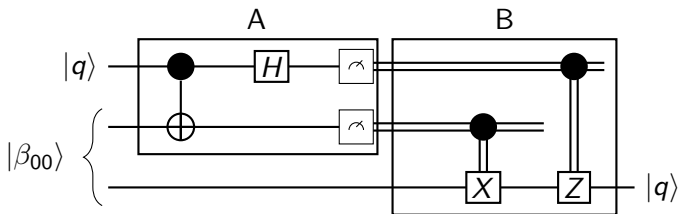
$$|00\rangle \xrightarrow{\text{SWAP}} |10\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle$$

$$\xrightarrow{\text{SWAP}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |00\rangle$$

$$|10\rangle \xrightarrow{\text{SWAP}} |00\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = |10\rangle$$

3.6 Quantum Teleportation

A communication protocol designed to transfer the information of a single qubit through a classical channel.



Instead of $|q\rangle$, A prepares the following state shared with B

$$\begin{aligned}
 |q\rangle |\beta_{00}\rangle &= \left(\alpha_0 (|0\rangle + |1\rangle) \right) \left(\frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \right) \\
 &= \frac{1}{\sqrt{2}} (\alpha_0 |000\rangle + \alpha_0 |011\rangle + \alpha_1 |100\rangle + \alpha_1 |111\rangle)
 \end{aligned}$$

$$\begin{aligned}
&\mapsto \frac{1}{\sqrt{2}}(\alpha_0 |000\rangle + \alpha_0 |011\rangle + \alpha_1 |110\rangle + \alpha_1 |101\rangle) \\
&\mapsto \frac{1}{\sqrt{2}}(\alpha_0 | +00\rangle + \alpha_0 | +11\rangle + \alpha_1 | -10\rangle + \alpha_1 | -01\rangle) \\
&= \frac{1}{2}(\alpha_0 |000\rangle + \alpha_0 |100\rangle + \alpha_0 |011\rangle + \alpha_0 |111\rangle \\
&\quad + \alpha_1 |010\rangle - \alpha_1 |110\rangle + \alpha_1 |001\rangle - \alpha_1 |101\rangle)
\end{aligned}$$

Suppose B measures the second qubit to be 0

$$\mapsto \frac{\sqrt{2}}{2}(\alpha_0 |00\rangle + \alpha_0 |10\rangle + \alpha_1 |01\rangle - \alpha_1 |11\rangle)$$

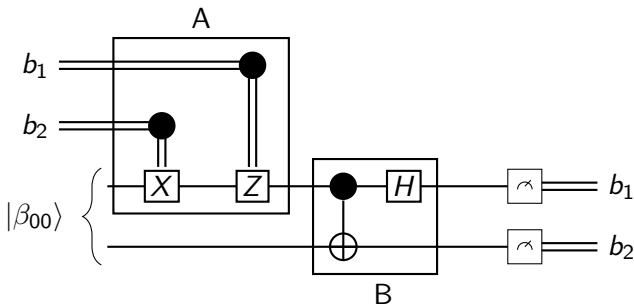
Finally, suppose it measures the first qubit to be 1

$$\mapsto \frac{\sqrt{2}\sqrt{2}}{2}(\alpha_0 |0\rangle + \alpha_1 |1\rangle) = |q\rangle$$

All other possible outcomes also yield $|q\rangle$.

3.7 Superdense Coding

A quantum communication protocol that transmits two classical bits of information using a single qubit.



It requires a pre-shared entangled pair and exploits entanglement to encode additional information through local operations.

It can be viewed as the reverse of quantum teleportation.

At the sender's side

$$\begin{aligned} |b_0\rangle &\rightarrow \frac{1}{\sqrt{2}}(|0\rangle\langle 0| \otimes I) + (I \otimes |0\rangle\langle 0|) \\ &\rightarrow \frac{1}{\sqrt{2}}((-1)^{b_0}|0\rangle\langle 0| + (-1)^{b_0+0}|0\rangle\langle 0|) = |b_0\rangle \end{aligned}$$

where we have used that $Z|b_0\rangle = (-1)^{b_0}|b_0\rangle$.

At the receiver's side

$$\begin{aligned} |b_0\rangle &\rightarrow \frac{1}{\sqrt{2}}((-1)^{b_0}|b_0\rangle\langle b_0| + (-1)^{b_0+0}|b_0\rangle\langle b_0|) \\ &\rightarrow \frac{1}{\sqrt{2}}((-1)^{b_0}|b_0\rangle + (-1)^{b_0+0}|b_0\rangle)|b_0\rangle \\ &\rightarrow (-1)^{b_0}Z|b_0\rangle = (-1)^{b_0+0}Z|b_0\rangle = (-1)^{b_0}I|b_0\rangle = |b_0\rangle \end{aligned}$$

$$= \frac{1}{2} \left[((-1)^{b_1 b_2} + (-1)^{b_1 \bar{b}_2}) |0\rangle + ((-1)^{b_1 b_2 + b_2} + (-1)^{b_1 \bar{b}_2 + \bar{b}_2}) |1\rangle \right] |b_2\rangle$$

Considering the two cases $b_1 = 0$ and $b_1 = 1$, we obtain

$$= |b_1\rangle |b_2\rangle$$

Measuring in the computational basis, the outcomes are b_1 and b_2 .

3.8 Role in Quantum Networks

Entanglement is not just a physical phenomenon — it is a communication resource.

- Quantum Teleportation

1. Transfers quantum states (not particles) using pre-shared entanglement and classical communication.
2. The original quantum state is destroyed upon measurement, ensuring *security*: no cloning or interception is possible.
3. Enables reliable state transfer across distant quantum nodes, forming a backbone of *distributed quantum computing*.
4. First experimentally demonstrated in 1997 by Zeilinger (Nobel Prize in Physics, 2022) and his team. Later extended to long distances via satellite-based links—notably China's *Micius* satellite (2017), achieving distances of up to 1,400 km—and to multi-node quantum networks.

- Superdense Coding

1. Transmits 2 classical bits by sending a single qubit, enabled by pre-shared entanglement.
2. Effectively doubles classical communication capacity relative to sending unentangled qubits.
3. Supports efficient classical information transfer and control signaling in quantum communication systems.
4. First experimentally demonstrated in 1996 by Zeilinger and his team. Practical deployment is limited by the difficulty of generating and distributing high-fidelity entanglement and performing efficient Bell-state measurements at scale.

Quantum teleportation and superdense coding form the dual pillars of the quantum internet and future financial infrastructure.

2.2 Strategic Relevance to Finance

2.2.1 Quantum Teleportation

1. **Prevents Interception or Duplication of Sensitive Quantum Information:** This ensures that proprietary models (e.g. risk models and simulations) can be transmitted without exposing their underlying structure.
2. **Supports cross-border access to quantum computing resources:** This enables the development of global quantum cloud infrastructures for financial institutions.

2.2.2 Superdense Coding

1. **Optimizes Bandwidth Utilization in Communication and Trading Networks:** This is critical for high-frequency trading and real-time data exchange.
2. **Enhances secure classical communication using entanglement-assisted protocols:** This strengthens confidentiality while improving the efficiency of data transmission between institutions.

4 Quantum Key Distribution

In contrast to Quantum Teleportation and Superdense Coding, which focus on the transfer of quantum states or classical information, the primary goal of QKD protocols like BB84 and E91 is secret key distribution.

4.1 Secret Key Distribution

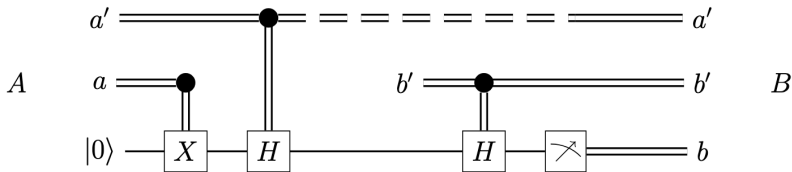
BB84 and E91 allow two parties to generate a shared secret key while detecting any eavesdropping during the exchange.

BB84 is the standard in most commercial implementations today, due to its simplicity, robustness, and compatibility with existing optical communication technologies.

E91 offers stronger theoretical security due to its reliance on quantum entanglement and Bell's inequalities. However, its implementations largely remain in pilot or advanced research stages.

4.2 BB84 Protocol

Suppose the two parties A and B agree on the following communication protocol



a and a' are binary bits obtained randomly and with equal probability from a trusted source.

The quantum state that exits the Hadamard gate of A is

$$|0\rangle \mapsto |0 \oplus a\rangle = |a\rangle \mapsto H^{a'}|a\rangle$$

Note that there are four possible states

$$\{a, a'\} = 00, 01, 10, 11 \mapsto |0\rangle, |+\rangle, |1\rangle, |-\rangle$$

On the other side of the quantum communication channel, B receives the qubit $H^{a'}|a\rangle$ and applies a controlled Hadamard gate depending on b'

$$H^{a'}|a\rangle \mapsto H^{b'}(H^{a'}|a\rangle) = H^{b'}H^{a'}|a\rangle$$

Since

$$H^{a'}|a\rangle = \left(\frac{1}{\sqrt{2}} \sum_j (-1)^{aj}\right)^{a'} |a\rangle$$

$$\begin{aligned} H^{b'}H^{a'}|a\rangle &= \left(\frac{1}{\sqrt{2}} \sum_i (-1)^{ai}\right)^{b'} \left(\frac{1}{\sqrt{2}} \sum_j (-1)^{aj}\right)^{a'} |a\rangle \\ &= \left(\frac{1}{\sqrt{2}} \sum_i (-1)^{ai}\right)^{b'+a'} |a\rangle = \left(\frac{1}{\sqrt{2}} \sum_i (-1)^{ai}\right)^{b' \oplus a'} |a\rangle \end{aligned}$$

There are eight possible cases: B does not apply H : $|0\rangle, |+\rangle, |1\rangle, |-\rangle$;
 B applies H : $H|0\rangle = |+\rangle, H|+\rangle = |0\rangle, H|1\rangle = |-\rangle, H|-\rangle = |1\rangle$.

The probability that B measures the state $|b\rangle$ is

$$P_B(|b\rangle) = \left| \langle b | \left(\frac{1}{\sqrt{2}} \sum_i (-1)^{ai} \right)^{b' \oplus a'} |a\rangle \right|^2 = \left| \left(\frac{1}{\sqrt{2}} \sum_i (-1)^{ai} \right)^{b' \oplus a'} \langle b | a \rangle \right|^2$$

If $a' = b'$

$$H^{a'} H^{a'} |a\rangle = |a\rangle$$

and

$$P_B(|b\rangle) = |\langle b | a \rangle|^2 = |\delta_{ab}|^2 = \delta_{ab}$$

But, if $a' \neq b'$

$$H^{b'} H^{a'} |a\rangle = H |a\rangle = |(-1)^a\rangle$$

and

$$P_B(|b\rangle) = \frac{1}{2}$$

In other words, $a' = b'$ implies that $a = b$, but $a' \neq b'$ implies that $a = b$ only 50% of the time.

Using a classical channel, A communicates to B the value of the random variable X .

If $X \neq Y$, the result is discarded, since B 's measurement does not reliably reflect A 's bit.

If $X = Y$, the measurement is consistent, and both parties keep the bit, agreeing that $Z = X$.

To generate a shared key of sufficient length, the procedure is repeated many times.

The security of the protocol relies on the fact that any third party (Eve) attempting to intercept the qubits must perform measurements, which inevitably disturb their quantum states.

As a result, any detected eavesdropper is exposed, and the protocol reveals the presence of an eavesdropper.

4.3 From Classical to Quantum Communication

Understanding how information is encoded in quantum systems is essential to understanding QKD.

From radio communication to optical communication systems (fiber optics, lasers), modern communication systems rely on *electromagnetic waves* to transmit information over long distances.

Predicted by Maxwell in 1864 and experimentally confirmed by Hertz in 1887. Electromagnetic waves include visible light (about a hundred times smaller than the thickness of a human hair).

Electric and magnetic fields oscillate in tandem and propagate through space, carrying energy and information from one point to another.

Light polarization is a controllable property that can be used to encode information. Light polarization is determined by the direction of the electric field of the electromagnetic wave.

Polarization is used in QKD at the level of single photons.

4.4 The Discrete Nature of Light

In quantum physics, light is composed of discrete packets of energy called *photons*.

The *photoelectric effect* — explained by Einstein in 1905, Nobel Prize in 1921 — demonstrated that light energy is quantized rather than continuous.

Each photon can carry information encoded in its quantum state — for example, through its polarization.

In QKD, information is transmitted at the level of individual photons rather than classical continuous signals.

This discrete, quantum nature is precisely what makes eavesdropping detectable: any attempt to intercept a photon disturbs its quantum state in a measurable way.

4.5 Light Polarization and Single Qubits

The polarization of a photon provides a natural way to encode quantum information.

The *polarization 1-qubit*

$$|\epsilon\rangle = \alpha_x |\epsilon_x\rangle + \alpha_y |\epsilon_y\rangle = \sum_{k=x,y} \alpha_k |k\rangle$$

Polarization states (horizontal, vertical, diagonal) form a two-level quantum system.

A photon can exist in a superposition of polarization states.

Measuring the polarization forces the photon into a definite state.

Measurement in the wrong basis introduces randomness and errors.

This measurement disturbance is the fundamental mechanism that enables secure key distribution in QKD.

4.6 One-Photon Polarization States

If the polarizer is rotated an angle θ relative to the x axis

$$|p_0\rangle = \cos \theta |\epsilon_0\rangle + \sin \theta |\epsilon_1\rangle$$

If it is rotated $\theta + 90^\circ$

$$|p_1\rangle = -\sin \theta |\epsilon_0\rangle + \cos \theta |\epsilon_1\rangle$$

In more compact notation

$$|p_i\rangle = \sum_j (-1)^{i \star j} \sin^{i \oplus j} \theta \cos^{i \oplus j \oplus 1} \theta |\epsilon_j\rangle$$

where

$$0 \star 0 = 0, \quad 0 \star 1 = 0, \quad 1 \star 0 = 1, \quad 1 \star 1 = 0$$

The probability of measuring the photon $|c\rangle$ in the state $|u\rangle$ is then

$$\begin{aligned} P(u) &= |\langle u|c\rangle|^2 = \left| \sum_j \alpha_j \langle u|c_j\rangle \right|^2 \\ &= \left| \sum_j \alpha_j \sum_k (1-i)^{k/2} \sin^{k/2} \theta \cos^{k/2} \theta \langle u|k\rangle \right|^2 \\ &= \left| \sum_j \alpha_j (-1)^{k/2} \sin^{k/2} \theta \cos^{k/2} \theta \right|^2 \end{aligned}$$

2.2 Polarized Polarization in Two-Qubit Systems

Let θ be the angle of rotation of the polarizer in the left. The left single qubit will be in the superposition state

$$|u\rangle = \sum_k (1-i)^{k/2} \sin^{k/2} \theta \cos^{k/2} \theta |k\rangle$$

If ϕ the angle of the polarizer on the right, the single qubit moving to the right will be

$$|p_j\rangle = \sum_{\ell} (-1)^{j \star \ell} \sin^{j \oplus \ell} \phi \cos^{j \oplus \ell \oplus 1} \phi |\epsilon_{\ell}\rangle$$

Given two-photon entangled state

$$|\epsilon_2\rangle = \sum_{m,n} \alpha_{mn} |\epsilon_m \epsilon_n\rangle$$

the probability of measuring it in $|p_i p_j\rangle$ is

$$\begin{aligned} P(|p_i p_j\rangle) &= |\langle p_i p_j | \epsilon_2 \rangle|^2 = \left| \langle p_i p_j | \sum_{m,n} \alpha_{mn} |\epsilon_m \epsilon_n\rangle \right|^2 \\ &= \left| \sum_{m,n} \alpha_{mn} \sum_{k,l} (-1)^{i \star k + j \star l} \sin^{i \oplus k} \theta \cos^{i \oplus k \oplus 1} \theta \sin^{j \oplus l} \phi \cos^{j \oplus l \oplus 1} \phi \delta_{km} \delta_{ln} \right|^2 \\ &= \left| \sum_{m,n} \alpha_{mn} (-1)^{i \star m + j \star n} \sin^{i \oplus m} \theta \cos^{i \oplus m \oplus 1} \theta \sin^{j \oplus n} \phi \cos^{j \oplus n \oplus 1} \phi \right|^2 \end{aligned}$$

4.8 Example: β_0

Substitute

$$|\beta_0\rangle = \frac{1}{\sqrt{2}}(|\epsilon_0 \epsilon_0\rangle + |\epsilon_1 \epsilon_1\rangle)$$

in the general formula above

$$\begin{aligned} P(|p_i p_j\rangle) &= |\langle p_i p_j | \beta_0 \rangle|^2 = \frac{1}{2} |\langle p_i p_j | (|\epsilon_0 \epsilon_0\rangle + |\epsilon_1 \epsilon_1\rangle)|^2 \\ &= \frac{1}{2} |(-1)^{i \star 0 + j \star 0} \sin^i \theta \cos^{i \oplus 1} \theta \sin^j \phi \cos^{j \oplus 1} \phi \\ &\quad + (-1)^{i \star 1 + j \star 1} \sin^{i \oplus 1} \theta \cos^i \theta \sin^{j \oplus 1} \phi \cos^j \phi|^2 \end{aligned}$$

In particular

$$P(|p_0 p_0\rangle) = P(|p_1 p_1\rangle) = \frac{1}{2} \cos^2(\theta - \phi)$$

$$P(|p_0 p_1\rangle) = P(|p_1 p_0\rangle) = \frac{1}{2} \sin^2(\theta - \phi)$$

Applying Bayes' theorem ($p_{r|s} = p_{r,s}/p_s$), the probability that B measures $|p_0\rangle$ given A measured $|p_0\rangle$ is

$$P(|\cdot p_0\rangle | |p_0 \cdot\rangle) = \frac{P(|p_0 p_0\rangle)}{P(|p_0 \cdot\rangle)}$$

For the EPR state $|\beta_0\rangle$

$$P(|\cdot p_0\rangle | |p_0 \cdot\rangle) = \frac{\frac{1}{2} \cos^2(\theta - \phi)}{\frac{1}{2}} = \cos^2(\theta - \phi)$$

If the two x polarizers are parallel $\theta - \phi = 0^\circ$

$$P(|\cdot p_0\rangle | |p_0 \cdot\rangle) = 1$$

The two photons are *maximally positive correlated*. When the two polarizers are perpendicular $\theta - \phi = 90^\circ$

$$P(|\cdot p_0\rangle | |p_0 \cdot\rangle) = 0$$

The two photons are *maximally negatively correlated*.

4.9 Unmasking the Silent Observer

The probability that E observes $|E_k\rangle$ given A measures $|p_i\rangle$ is $P(|p_i\rangle |E_k\rangle)$

$$\left| \sum_{m,n} \alpha_{mn} (-1)^{i \oplus m + k \oplus n} \sin^{i \oplus m} \theta \cos^{i \oplus m \oplus 1} \theta \sin^{k \oplus n} \varphi \cos^{k \oplus n \oplus 1} \varphi \right|^2$$

where φ is the angle that E 's polarizer makes with the x -axis.

Assume that each time A measures $|p_i\rangle$ and E observes $|E_k\rangle$, E transmits $|E_k\rangle$ to B . The two-photon polarization state prior to measurement by B is

$$|p_i\rangle |E\rangle = \sum_k \sqrt{P(|p_i\rangle |E_k\rangle)} |p_i\rangle |E_k\rangle$$

The probability that B observes $|p_j\rangle$ is then

$$P_E(|p_i\rangle |p_j\rangle) = |\langle p_i | p_j \rangle \langle p_i | E \rangle|^2 = \left| \sum_k \sqrt{P(|p_i\rangle |E_k\rangle)} \langle p_j | E_k \rangle \right|^2$$

Consider the case where the polarizers of E and F are aligned ($j = k$)

$$P_E(n, n) = \left| \sum_j \sqrt{P(n, E_j)} (-1)^{n+j} e^{i\phi_j} e^{i\phi_j} \right|^2$$

If, however, $|E_k\rangle = |j\rangle$, and $j = k$, then

$$\begin{aligned} P_E(n, n) &= \left| \sum_j \sqrt{P(n, E_j)} (-1)^{n+j} e^{i\phi_j} e^{i\phi_j} \right|^2 \\ &= \left| \sqrt{P(n, E_n)} \right|^2 = P(n, n) \end{aligned}$$

Thus,

$$P_E(n, n) = P(n, n)$$

only if E aligns its polarizer with F and performs the same measurement.
While not impossible, this event is extremely unlikely: the statistical
probability of multiple or more photons getting aligned.

4.10 E91 Protocol Example

1. A and B share the EPR state $|\beta_0\rangle$ and independently choose angles θ, ϕ from $\{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$.
2. Photon pairs are detected based on their local, private orientation. Measurements at 45° and 135° are reserved for testing Bell's inequalities.
3. After verifying channel security, they publicly announce their orientations to “sift” the data.
4. Correlated outcomes ($P = 1/2$) from aligned bases generate the secret key

$$\begin{aligned}\text{Parallel } (\theta = \phi) : \quad & P(|p_0; p_0\rangle) = P(|p_1; p_1\rangle) \implies \text{bit } i \\ \text{Perpendicular } (\theta \perp \phi) : \quad & P(|p_0; p_1\rangle) = P(|p_1; p_0\rangle) \implies \text{bit } \bar{i}\end{aligned}$$

This key generation process, protected by quantum correlations, enables confidential communication with complete secrecy.

4.10 Continuous Variables

Quantum information can be encoded not only in discrete systems, such as the polarization of single photons, but also in continuous variables. These systems provide an alternative framework for quantum communication, including QKD.

The theoretical description of these systems is based on the quantization of the electromagnetic field, as studied in *quantum optics*.

Coherent states of light have minimum uncertainty and remain stable under propagation. They most closely resemble classical coherent light (eg laser beams).

Reminder: *Heisenberg's uncertainty principle*: $\Delta x \cdot \Delta p \geq \hbar/2$.

In *squeezed states*, the uncertainty is reduced in one variable at the expense of increased uncertainty in the conjugate variable. This allows noise reduction and improved measurement precision.

Measurement precision can surpass classical limits.

5 Post-Quantum Security Measures and Real-World Deployment

As quantum hardware advances, financial institutions face a concrete and time-sensitive transition challenge: securing critical infrastructure against both present vulnerabilities and future quantum-enabled threats.

Transitioning to a quantum-secure environment requires an institution-wide strategy that integrates security into daily operations.

This roadmap begins with operational risk identification and readiness audits to inventory vulnerabilities. These findings are embedded into risk management frameworks, guiding critical infrastructure modernization.

Crucially, organizations must align governance and compliance with evolving standards, build necessary workforce capability, and implement quantum-secured interconnectivity to ensure long-term resilience against future decryption threats.

5.1 Operational Risk Identification

A robust quantum security strategy begins by *assessing operational risk*.

Financial institutions must identify emerging attack surfaces by mapping all systems relying on quantum-vulnerable algorithms (eg RSA, ECC) for migration.

Data with long-term confidentiality requirements is immediately vulnerable to the HNDL threat model:

Critical data: High risk; requires immediate PQC or QKD implementation.

Confidential data: Moderate risk; subject to lingering HNDL exposure.

Cryptographic vulnerabilities in vendor systems and external channels compound institutional risk and must be integrated into comprehensive audits.

Risk assessment must be continuous, evolving alongside hardware capabilities and threat timelines.

5.2 Quantum Audits

While risk identification assesses the threat landscape, readiness audits evaluate infrastructure maturity.

They transition the strategy from abstract risk prioritization to concrete technical assessment.

The audit begins by cataloging cryptographic dependencies across all systems and workflows—inventorizing every algorithm, library, and communication channel currently in use.

Assets are then subjected to a gap analysis, measuring the distance between current infrastructure and target quantum-safe standards, such as NIST PQC guidelines. This technical baseline converts high-level risks into a migration roadmap.

A quantum readiness audit ensures that subsequent governance and remediation decisions are driven by formal system inventories rather than speculation.

5.3 Risk Management Framework

Quantum risk must not be managed in isolation; it requires integration into established enterprise risk management (ERM) structures to ensure institutional sustainability.

Insights from risk identification and readiness audits must be embedded within ERM, defining clear ownership, priorities, and mitigation paths for all identified vulnerabilities.

Institutional resilience should be stress-tested against defined quantum threat scenarios, modeling both accelerated hardware timelines and worst-case cryptographic compromise.

The objective is not to eliminate quantum risk entirely, but to govern it in a structured, measurable, and risk-proportionate manner that aligns with overall institutional strategy.

5.4 Critical Infrastructure Modernization

Translating readiness audits and risk frameworks into action requires the systematic migration of cryptographic infrastructure toward quantum-secure standards.

Institutions must execute a prioritized, time-bound roadmap replacing vulnerable algorithms with NIST-approved PQC schemes, specifically ML-KEM, ML-DSA, and SLH-DSA.

To maintain operational stability, deploy a *hybrid transition*—running classical and post-quantum algorithms in parallel. This ensures interoperability while mitigating service disruption.

Finally, prioritize *cryptographic agility*. Future systems must be architected to allow the swapping of cryptographic primitives without full infrastructure rebuilds, ensuring long-term adaptability as standards evolve.

Migration is a multi-year strategic imperative; institutional resilience depends on the discipline of this transition.

3.3 Governance and Compliance

Regulatory landscapes for quantum security are evolving rapidly. Organizations must align internal policies with global standards—including NIST, NSA/CSSA 2.0, and EU PQ/C roadmaps—proactively rather than waiting for compliance to become mandatory.

Quantum security demands fiduciary oversight. This requires visible governance, with clear accountability assigned to risk committees and regular reporting to senior leadership.

Internal policies must be codified across the full supply chain, updating vendor procurement standards, internal cryptographic policies, and third-party risk assessments to maintain quantum-safe readiness.

Transparency is the final requirement: that quantum readiness efforts, investments are sustained, prioritized, and enforceable across the entire organization.

5.6 2026 Quantum Cryptographic Transition Roadmap

Phase	Duration	Focus
Phase 1: Scope	Weeks 1-4	Identify the 20% of systems that carry 80% of the PQC risk.
Phase 2: Auto-Discover	Weeks 5-10	Run automated scans to map crypto-usage on that 20% (Crown Jewels).
Phase 3: Assess/Gap	Weeks 11-20	Map usage to NIST standards; identify the "Quantum Gaps."
Phase 4: Roadmap	Weeks 21-28	Develop the remediation schedule (for 2027+).
Phase 5: Document	Weeks 29-32	Compile the "Compliance Report" for regulators.

5.2 Workforce Capability

Technology alone is insufficient. Institutions must build the workforce capabilities needed to harness quantum, blockchain, and other disruptive technologies over time.

Policy, regulatory, and technology leaders must understand the opportunities and challenges that quantum technology presents and develop strategies to maximize benefits and minimize risks.

Proper technical capability in quantum computing, cryptography, quantum communications, and other quantum systems must be cultivated through targeted programs and external partnerships.

IT, HR, legal, and business functions must be aligned around a shared quantum security roadmap, with clear roles and responsibilities at each stage of the transition.

The quantum transition is as much an organizational challenge as a technical one. Workforce capability determines whether institutions can harness their quantum technology effectively.